

Final Year BTP — Diabetes Risk Prediction

Final Project Documentation and Summary

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Project Overview

This project implements a complete end-to-end machine learning pipeline using the Pima Indians Diabetes dataset. It covers dataset ingestion, cleaning, preprocessing, model training, explainable AI with SHAP, and final deployment via Streamlit.

Pipeline Summary

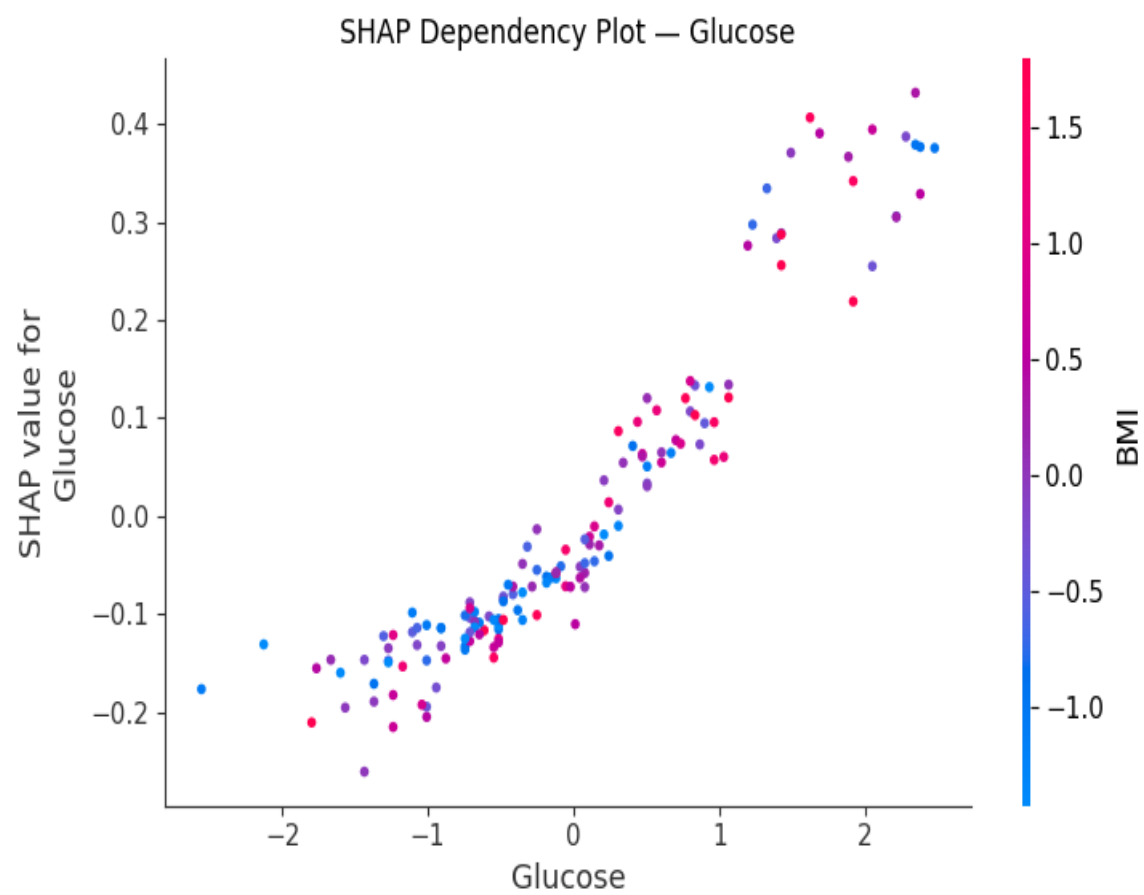
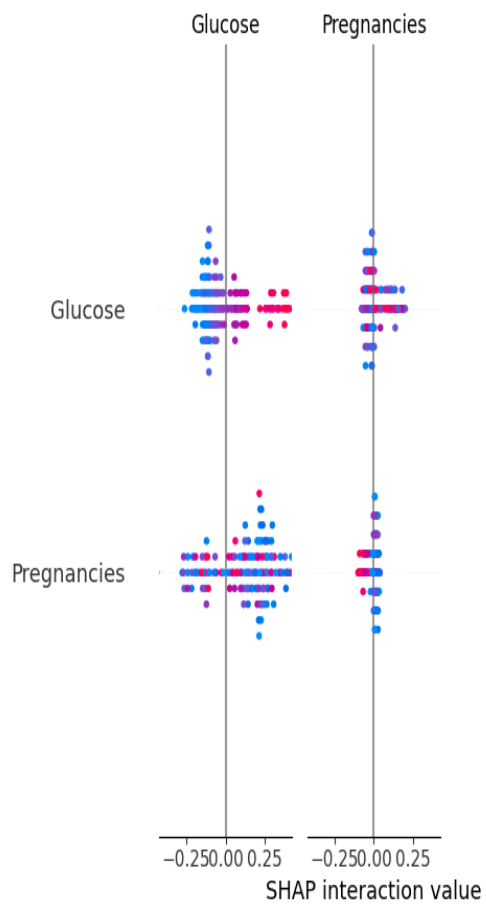
1. **Setup & EDA:** Environment setup, dataset inspection, and correlation analysis.
2. **Data Cleaning:** Replaced biologically impossible zeros with NaN, imputed missing values, and standardized features.
3. **Model Training:** Trained Logistic Regression, Random Forest, XGBoost, and LightGBM models.
4. **Explainability:** Used SHAP for global and local interpretations.
5. **Automated Reports:** Generated detailed PDF & HTML reports with plots.
6. **Interactive Interface:** Built an interactive Colab/Streamlit app for predictions.
7. **Deployment:** Packaged and deployed the app publicly for demonstration.

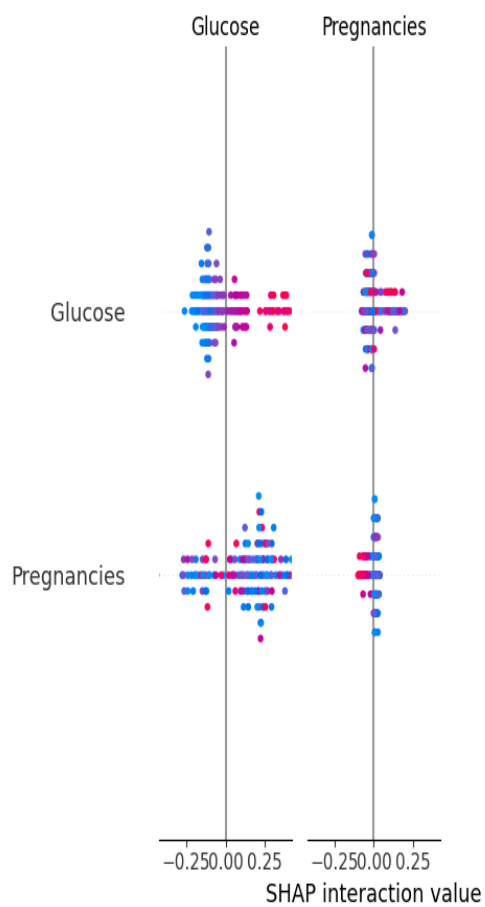
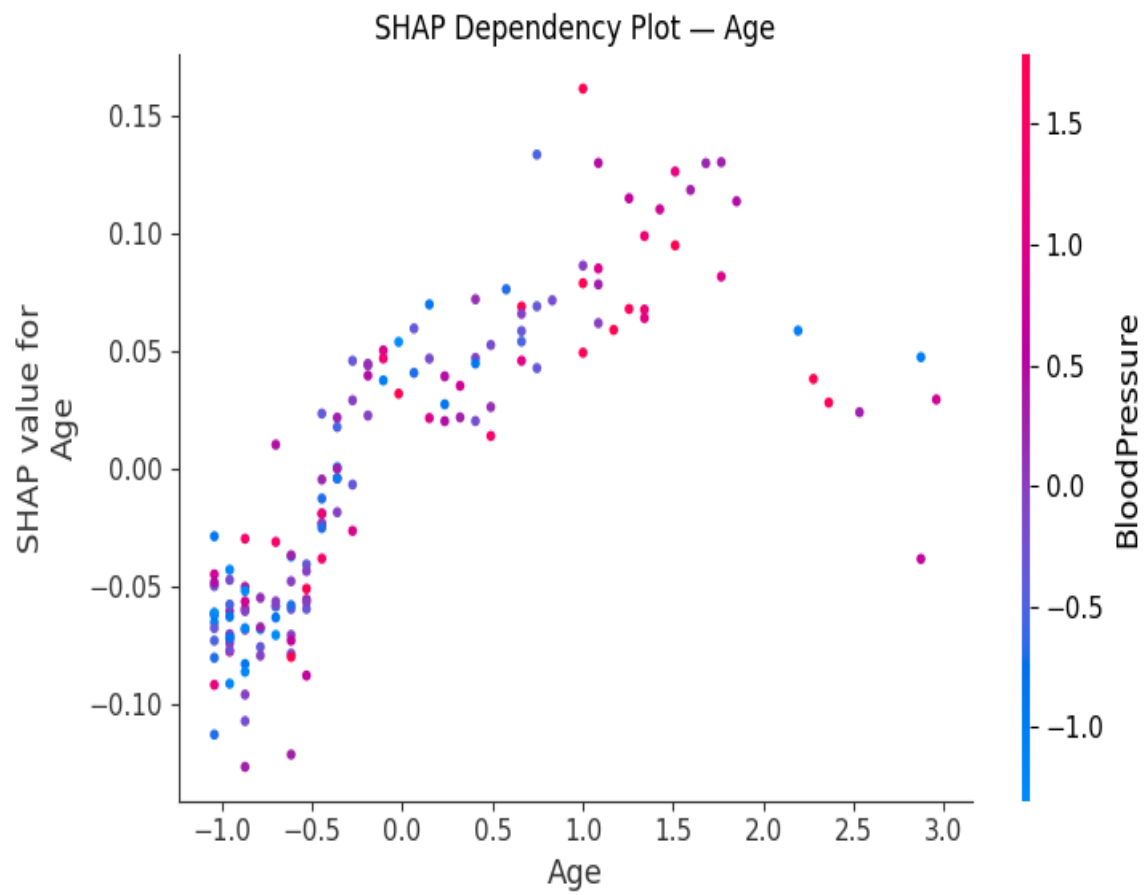
Key Findings

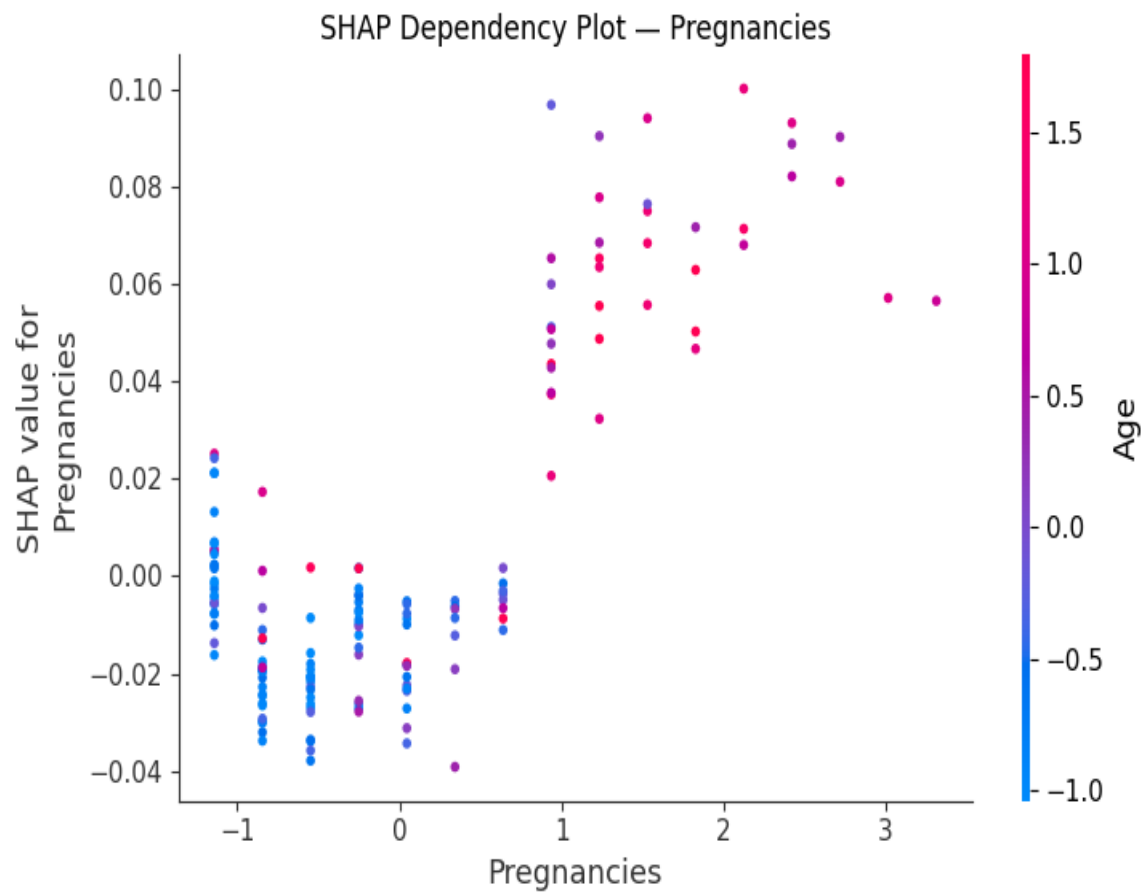
- The best model by F1-score: **XGBoost**.
- **Note:** Due to technical issues encountered with converting the base value score for XGBoost for SHAP analysis, the **Random Forest** model was used for generating SHAP explanations instead.
- Dominant predictors: **Glucose, BMI, Age, and Pregnancies**.
- SHAP analysis produced interpretable feature contributions for each prediction.

Files Generated

- /data/processed/ → Cleaned dataset & preprocessor
- /results/ → Trained models & metrics
- /explain/ → SHAP plots & interpretations
- /reports/ → Auto-generated PDF & dashboards
- /docs/ → Text summaries
- app.py → Streamlit deployment app







Conclusion

This workflow demonstrates a reproducible, explainable, and deployable ML solution for early diabetes risk detection. The project integrates robust preprocessing, multi-model evaluation, SHAP explainability, and an interactive prediction interface — producing a comprehensive, real-world deployable framework.